

Emerging trends in cyber security

The walls that used to protect a network have basically dissolved. This dot point is about the five ideas the whole industry is shifting toward — and why "keep the bad guys outside" is no longer a plan.

What this key knowledge covers

This summary distils everything you need for **KK01 — Emerging trends in cyber security** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

The castle had one wall. That wall is gone.

For decades, network security worked like a medieval castle. You built one big wall — a *firewall* — around your network. Anyone **outside** was a stranger to be checked. Anyone **inside** was trusted. Get past the wall once, and you could wander freely.

That model quietly broke. People now work from home, from phones, from cafés. Data lives in the cloud, not in a server room you can lock. Clutch's instructors log in from their cars; its database sits on rented cloud servers in another state. **There is no longer a single "inside" to defend.** So the field has shifted from *perimeter security* ("guard the wall") to a set of newer ideas that assume the attacker is already in, and check everything, all the time.

VCAA lists these under "**emerging trends** in cyber security." When a question asks you to *describe* or *explain* a trend, it wants two things: **what it is**, and **what problem it solves** compared with the old approach. A definition alone rarely gets full marks.

One job each

Each trend attacks a different weak spot. Read them as a set: together they replace the single castle wall with many small, constant checks.

Proving who you are with more than just a password.

Never trust, always verify — even inside the network.

Machine learning spots the attack a fixed rule would miss.

Scrambling data so stolen files are useless.

Laws forcing organisations to actually do the above.

Multi-factor authentication

Multi-factor authentication (MFA) means proving your identity using **two or more independent factors** from different categories. The three categories are easy to remember:

The whole point is **independence**. An attacker in another country can buy a leaked Clutch password, but they don't have the student's *phone*. That's why the second factor stops them. Try the breach below.

An attacker has phished an instructor's password. Watch what each defence does when the password falls.

Zero Trust Architecture

Zero Trust Architecture (ZTA) throws out the idea of a trusted "inside." Its motto is "never trust, always verify." Every request — even from a device already on the network — must prove who it is, and gets only the **minimum access** it needs (*least privilege*). Crucially, the network is broken into small zones (*micro-segmentation*) so a break-in can't spread.

The scene below is the heart of this dot point. Compromise one machine and watch what happens under each model. In **Castle & Moat**, one breach owns everything. In **Zero Trust**, every hop hits a checkpoint, so the damage is contained.

Pick a model, then click a node. The breach tries to spread to every connected machine.

Never trust, always verify (re-check every request) · **Least privilege** (give the smallest access that still works — a Clutch instructor sees *their* learners, never the whole database) · **Micro-segmentation** (wall the network into zones so one breach is trapped). Name these three and you've covered the dot point.

AI threat detection

Old defences used **fixed rules**: "block any login from outside Australia," "alert if more than 5 failed passwords." Attackers learn the rules and slip under them. *AI threat detection* instead uses **machine learning** to learn what *normal* traffic looks like, then flags anything **anomalous** — even a pattern no one wrote a rule for.

On Clutch, a real instructor opens a handful of student profiles a day. The graph below shows one account's profile-views per minute. At 14:30 it spikes — someone is scraping every learner's home address. A fixed threshold set at 50 never trips. The model, which learned this account's normal rhythm, flags it instantly.

AI detection catches **novel, unknown** attacks and adapts as behaviour changes — but it can raise **false positives** (flagging harmless oddities) and needs lots of clean data to learn from. A good answer names both the strength *and* a limitation.

Cryptographic protection

Cryptography scrambles readable data (*plaintext*) into unreadable *ciphertext* using a **key**. Without the key, a stolen file is gibberish. Modern systems encrypt data in two places: **in transit** (moving across the network, e.g. HTTPS) and **at rest** (sitting in the database). If an attacker stole Clutch's database but the permit numbers were encrypted, they'd get nothing usable.

The simplest cipher is the **Caesar cipher** — shift every letter by a fixed amount. Play with it to feel how a key works, then read why this exact idea is hopeless today.

Regulatory compliance

The first four trends are *technical*. This one is the **reason organisations bother**. *Regulatory compliance* means following the laws and standards that govern how data must be protected. It turns "we should secure data" into "we are legally required to." Because Clutch stores minors' personal data and location, several Australian rules apply:

Set the baseline: only collect data you need, keep it secure (APP 11), and let people access their own data. Clutch must protect those permit numbers and addresses by law.

If a breach is likely to cause serious harm, Clutch **must** notify affected users and the Information Commissioner. Hiding a breach is itself illegal.

A government baseline of eight practical strategies — patching, MFA, restricting admin rights, backups — that organisations are assessed against. MFA from Trend 01 is one of the eight.

Governs how Victorian public bodies handle personal data; relevant if Clutch ever partners with VicRoads or a public school.

Compliance is an *emerging* trend because regulation keeps tightening — the NDB scheme is recent, fines have grown, and "we didn't know" is no longer a defence. Security is shifting from optional to mandatory.

Five analogies that stick

An ATM needs your **card** (have) *and* your **PIN** (know). Steal one, you still can't withdraw. That's two factors from two categories — textbook MFA.

Inside the terminal you're still checked at every gate, lounge and boarding pass scan. Being "past security" once doesn't let you board any plane. Verify every time.

Your bank texts "was this you?" after an odd purchase. It learned *your* normal spending, so an unusual one stands out — no fixed rule needed. That's anomaly detection.

A diary written in a code only you can read is useless if stolen. Encryption is that code — the data is there, but meaningless without the key.

Buildings have smoke alarms because the law demands it, not just because it's nice. Compliance makes good security non-optional — with penalties if you skip it.

Command-word decoder

VCAA marks against the **command word**. Same topic, different verb, completely different answer. Match your effort to the word.

Name it. "Multi-factor authentication." No explanation expected.

Say what it is *and* a defining feature. "MFA requires two or more factors from different categories, e.g. a password and a phone code."

Say *how/why* it works. Show cause and effect — why the trend solves the problem.

Give both sides, on the same criteria. "Unlike a firewall, Zero Trust verifies inside traffic too..."

Make a choice and defend it with reasons tied to the scenario. Link to Clutch's actual data and users.

Weigh strengths against limitations, then give a judgement. Both sides, then a verdict.

P-E-L writing trainer

VCAA answers reward a simple shape: **Point** (name it) → **Explain** (how/why it works) → **Link** (tie it to the scenario). Write a response below — the keyword meter lights up as you hit the ideas examiners look for.

Under Zero Trust, the compromised laptop is **not automatically trusted** just because it's on the network — every request it makes must still be **verified** (Point/Explain). Because of **least privilege**, that instructor account can only reach *its own* learners' records, not the full database, and **micro-segmentation** walls the network into zones so the attacker **can't move laterally** to other systems (Explain). This means the breach is contained to one instructor's small set of Clutch data rather than exposing all 40,000 users' permit numbers and home addresses (Link).

VCE-style practice questions

Five questions, **20 marks** total, building from *Identify* to *Evaluate*. Each shows a weak and a strong response, a marking guide, then mark your own — your score feeds the dial in the corner.

"It's when you use more than one password to log in so it's more secure."

"MFA requires a user to verify their identity with two or more factors from different categories — something you know (password), something you have (a phone code) and/or something you are (a fingerprint)."

- 1 mark: two or more factors required to authenticate.
- 1 mark: factors come from *different categories* (know / have / are), ideally with an example.

"Zero Trust is more secure because it doesn't trust anyone, so hackers can't get in. A firewall is old technology."

"A firewall only checks traffic at the network edge, so once an attacker is inside — say via a stolen instructor login — they're trusted and can roam. 'Never trust, always verify' re-checks every request and grants least privilege, so a compromised instructor account can only reach its own learners, not Clutch's whole database of 40,000 permit numbers and addresses."

- 1 mark: firewall trusts everything once inside the perimeter.
- 1 mark: Zero Trust verifies each request / applies least privilege.
- 1 mark: linked to Clutch's data being contained / not fully exposed.

"AI is better than rules because AI is smart and modern. Rules are simple and don't work as well."

Quick quiz

Six rapid questions to test recall. Pick an answer to see why it's right or wrong.

If you remember nothing else

The field moved from "**guard the wall**" (perimeter) to "**verify everything, all the time, and assume you'll be breached**". Every trend is one piece of that shift. In a case study: **name the trend, explain the mechanism, link it to the specific data and users**.

Key terms

Term	Meaning
Something you know	A password, PIN or passphrase. Weakness: it can be guessed, phished, leaked or reused.
Something you have	A phone receiving a code, an authenticator app, a physical security key. Weakness: it can be lost or stolen — but not from across the internet.
Something you are	A fingerprint, face or voice — biometrics . Weakness: it can't be changed if it ever leaks.
Multi-factor authentication	Two+ factors from different categories (know / have / are). Stops a stolen password alone.
Zero Trust Architecture	Never trust, always verify + least privilege + micro-segmentation. Contains a breach.
AI threat detection	ML learns normal traffic, flags anomalies. Catches novel attacks; risks false positives.
Cryptographic protection	Encrypts data in transit & at rest. Stolen data is unreadable without the key. AES (symmetric), RSA (asymmetric).
Regulatory compliance	Privacy Act + APPs, NDB scheme, Essential Eight, PDP Act. Makes security mandatory.

Common traps & misconceptions

Exam trap

⚠ Exam trap A password **and** a security question is not MFA — both are "something you know." MFA needs factors from **different categories**. Examiners deliberately test this.

Exam trap

⚠ Exam trap Zero Trust is **not a product you install** and it's not "a really strong firewall." It's an **architecture / approach** built from verification, least privilege and segmentation. Calling it "better antivirus" loses the mark.

Exam trap

⚠ Exam trap A Caesar cipher has only **25 possible keys** — a computer cracks it instantly by trying them all (a *brute-force* attack). Real encryption like **AES-256** has more keys than there are atoms in the observable universe. Don't confuse "a cipher" with "secure encryption."

Exam trap

⚠ Trap 1 · "MFA = two passwords" Two things you *know* is one factor twice. MFA needs different **categories**: know + have + are.

Exam trap

⚠ Trap 2 · "Zero Trust is a firewall" It's an **architecture** — verify everything, least privilege, micro-segmentation. A firewall is just one component.

Exam trap

⚠ Trap 3 · "AI detection is always right" Mention a limitation — **false positives** and the need for training data — or you'll miss the evaluate marks.

Exam trap

⚠ Trap 4 · "Encryption stops hackers getting in" Encryption doesn't block access — it makes stolen data **unreadable**. It protects data, not the door.

Exam trap

⚠ Trap 5 · Listing without linking In a case study, naming the trend isn't enough. Tie it to the scenario — *which* Clutch data, *which* user, *which* threat.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Describe · 2 marks

What is meant by multi-factor authentication.

✗ A weak answer

"It's when you use more than one password to log in so it's more secure." Wrong — two passwords are the same factor. No mention of independent categories. ½/2.

✓ A strong answer

"MFA requires a user to verify their identity with two or more factors from different categories — something you know (password), something you have (a phone code) and/or something you are (a fingerprint)." Defines it and names the distinct categories with examples. 2/2.

Q2. Explain · 3 marks

Clutch currently relies on a single firewall. Explain why "never trust, always verify" offers better protection for its users' data.

✗ A weak answer

"Zero Trust is more secure because it doesn't trust anyone, so hackers can't get in. A firewall is old technology." Vague, no mechanism, and "can't get in" is wrong. Not linked to Clutch's data. 1/3.

✓ A strong answer

"A firewall only checks traffic at the network edge, so once an attacker is inside — say via a stolen instructor login — they're trusted and can roam. 'Never trust, always verify' re-checks every request and grants least privilege, so a compromised instructor account can only reach its own learners, not Clutch's whole database of 40,000 permit numbers and addresses." Mechanism, contrast with the firewall, and a scenario link. 3/3.

Q3. Compare · 4 marks

AI-based threat detection with traditional rule-based detection for spotting unusual activity on the Clutch network.

✗ A weak answer

"AI is better than rules because AI is smart and modern. Rules are simple and don't work as well." Asserts superiority without criteria; no real comparison; ignores AI's downsides. 1/4.

✓ A strong answer

"Rule-based detection uses fixed thresholds (e.g. alert after 50 profile views), so it's predictable and cheap but misses any attack designed to stay under the limit, like a slow scrape of learner addresses. AI detection learns each account's normal behaviour and flags anomalies even without a matching rule, catching novel attacks — but it can produce false positives and needs large amounts of clean training data. AI adapts; rules don't, but rules are simpler to audit." Same criteria both ways, strengths and limitations, scenario-linked. 4/4.

Q4. Recommend · 5 marks

Clutch's database of permit numbers and addresses is stored unencrypted. Recommend and justify **two** emerging-trend strategies to better protect this data.

✗ A weak answer

"They should use encryption and a strong password. That will keep hackers out and make it safe." One real strategy, one weak generic point; no justification tied to consequences. 2/5.

✓ A strong answer

"First, encrypt the data at rest with AES so that if the database is stolen, the permit numbers and addresses are unreadable without the key — turning a catastrophic breach into a non-event. Second, enforce MFA on all instructor and admin accounts, so a phished password alone can't unlock access to that data. Together they protect the data on two fronts: encryption limits the impact of a breach, while MFA reduces the chance of one. Given Clutch holds minors' home addresses, this also helps meet its obligations under the Privacy Act (APP 11)." Two valid strategies, each justified by consequence, plus a compliance link. 5/5.